Pharmaceutical Product Data Analysis for Price Optimization & Drug Comparison

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1. Abstract

The pharmaceutical industry generates a massive amount of medicine-related data involving drug pricing, formulations, pack sizes, compositions, and manufacturers. Understanding these relationships is essential for healthcare analytics, pharmaceutical retail management, market segmentation, and price optimization.

This project focuses on Exploratory Data Analysis (EDA) of pharmaceutical product data collected from OpenDataBay. The dataset contains over 2.5 lakh pharmaceutical products with information about medicine names, prices, compositions, pack sizes, manufacturers, and medicine categories.

The analysis was performed using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn. Multiple preprocessing techniques including missing value handling, feature extraction, outlier detection, and visualization were applied to analyze pricing behavior and pharmaceutical market trends.

Through univariate, bivariate, and multivariate analysis, the project identified major pricing patterns across medicine forms, manufacturers, strengths, and packaging types. The analysis revealed that tablets dominate the affordable medicine market, while injections contribute significantly to premium pricing and outliers.

The findings from this project can support pharmaceutical price optimization, drug comparison systems, healthcare analytics, and pharmaceutical market segmentation.

2. Introduction

2.1 Background

The pharmaceutical industry plays an important role in healthcare by manufacturing and distributing medicines for different diseases and treatments. Pharmaceutical products vary based on:

- Medicine form
- Drug composition
- Dosage strength
- Packaging size
- Manufacturer
- Therapeutic category

These factors create significant variations in pharmaceutical pricing.

With the availability of healthcare datasets, data analytics and visualization techniques can be used to identify hidden trends, pricing patterns, and medicine distribution behavior. Exploratory Data Analysis (EDA) helps understand pharmaceutical product characteristics and provides insights into market behavior.

2.2 Problem Statement

The pharmaceutical market contains medicines ranging from low-cost generic tablets to extremely expensive specialty injections. Understanding these pricing differences is important for:

- Drug comparison
- Healthcare affordability analysis
- Pharmaceutical market segmentation
- Price optimization
- Manufacturer-level pricing analysis
- Product positioning

This project aims to analyze pharmaceutical product data and identify how different factors influence medicine pricing.

2.3 Objectives

The major objectives of this project are:

1. To analyze pharmaceutical pricing behavior.
2. To identify the relationship between medicine form and price.
3. To compare pharmaceutical manufacturers based on pricing.
4. To analyze pack size influence on price.
5. To identify high-cost and low-cost medicines.
6. To study the impact of composition strength on pricing.
7. To perform univariate, bivariate, and multivariate analysis.
8. To identify outliers and pharmaceutical market trends.

3. Dataset Collection and Description

3.1 Dataset Source

- Source: OpenDataBay.com
- Dataset Name: Indian Medicine Dataset
- File Format: CSV

3.2 Dataset Overview

Attribute	Details
Total Rows	253,973
Total Columns	9
Dataset Type	Structured Tabular Data
Domain	Pharmaceutical Product Analytics

3.3 Dataset Columns

Column Name	Description
id	Unique identifier
name	Medicine name
price(₹)	Medicine price
Is_discontinued	Whether the medicine is discontinued
manufacturer_name	Manufacturing company
type	Allopathy / Non-allopathy
pack_size_label	Packaging information
short_composition1	Primary active ingredient
short_composition2	Secondary active ingredient

3.4 Sample Dataset Records

Medicine Name	Price	Manufacturer	Pack Size	Composition
Augmentin 625 Duo Tablet	₹223.42	GlaxoSmithKline Pharmaceuticals Ltd	Strip of 10 tablets	Amoxycillin + Clavulanic Acid
Azithral 500 Tablet	₹132.36	Alembic Pharmaceuticals Ltd	Strip of 5 tablets	Azithromycin
Ascoril LS Syrup	₹118.00	Glenmark Pharmaceuticals Ltd	Bottle of 100 ml Syrup	Ambroxol + Levosalbutamol
Allegra 120 Tablet	₹218.81	Sanofi India Ltd	Strip of 10 tablets	Fexofenadine

4. Technologies and Tools Used

Technology	Purpose
Python	Programming language
Pandas	Data preprocessing
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical plotting
Scikit-learn	Encoding and preprocessing
Jupyter Notebook	Development environment

5. Data Understanding

5.1 Understanding Dataset Structure

Before performing analysis, understanding the dataset structure is essential.

df.info()

Used to understand:

- Number of rows and columns
- Data types
- Missing values
- Column information

Code

```
df.info()
```

df.describe()

Used to analyze numerical columns.

Provides:

- Mean
- Median
- Standard deviation
- Minimum value
- Maximum value
- Quartiles

Code

```
df.describe()
```

5.2 Data Types

Numerical Feature

- price(₹)

Categorical Features

- name
- manufacturer_name
- type
- pack_size_label
- short_composition1
- short_composition2

5.3 Numerical and Categorical Columns

Numerical Columns

Numerical columns contain measurable values.

Example:

- price(₹)

Categorical Columns

Categorical columns contain labels or categories.

Examples:

- manufacturer_name
- pack_size_label
- medicine name
- composition columns

5.4 Statistical Analysis

Initial Statistics of Price Column

Metric	Value
Minimum Price	₹0
Maximum Price	₹4,36,000
Mean Price	₹270
Median Price	₹79
Distribution Type	Right-Skewed

Interpretation

- Most medicines fall under affordable pricing.
- A few medicines create extreme outliers.
- Mean price is highly affected by injectio

5.5 Choosing Visualization Techniques

Different visualizations are used depending on the data type.

Data Type	Visualization	Purpose
Numerical Data	Histogram	Distribution analysis
Numerical Data	Boxplot	Outlier detection
Numerical Data	Violin Plot	Density and spread
Categorical Data	Count Plot	Frequency analysis
Categorical Data	Bar Plot	Category comparison
Categorical Data	Pie Chart	Percentage distribution
Mixed Data	GroupBy + Barplot	Relationship analysis
Multiple Features	Heatmap	Correlation analysis

6. Data Preprocessing

6.1 Handling Missing Values

The short_composition2 column contained many missing values because several medicines only contained one active ingredient.

Code

```
df['short_composition2'] = df['short_composition2'].fillna("None")
```

Result

- Missing values were handled successfully.
- Dataset consistency improved.
- Analysis became easier.

6.2 Outlier Detection and Removal

The medicine price column contained extremely large outliers caused mainly by specialty injections.

Why Remove Outliers?

Outliers:

- Distort visualizations
- Affect average values
- Increase skewness
- Reduce analysis clarity

IQR Method

Code

```
Q1 = df['price(₹)'].quantile(0.25)
Q3 = df['price(₹)'].quantile(0.75)
IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

df = df[(df['price(₹)'] >= lower_limit) &
        (df['price(₹)'] <= upper_limit)]
```

Observations

- Most outliers belonged to injections.
- Outliers were valid medicines.
- Cleaned data improved visualization clarity.

7. Feature Engineering and Feature Extraction

7.1 Medicine Form Extraction

Medicine forms such as:

- Tablets
- Syrups
- Creams
- Suspensions
- Injections

were extracted from medicine names and pack size labels.

Purpose

This enabled:

- Form-wise analysis
- Form-wise visualization
- Average price comparison
- Percentage distribution analysis

7.2 Strength Extraction

Strengths such as:

- 500mg
- 10mg
- 25mg
- 100ml

were extracted from composition columns.

Purpose

- Strength vs price analysis
- Dosage comparison
- Correlation analysis

7.3 Label Encoding

Categorical features were converted into numerical format using Label Encoding.

Purpose

- Correlation analysis
- Heatmap generation
- Multivariate analysis

8. Exploratory Data Analysis (EDA)

8.1 Univariate Analysis

Univariate analysis studies a single variable individually.

8.1.1 Histogram of Medicine Prices

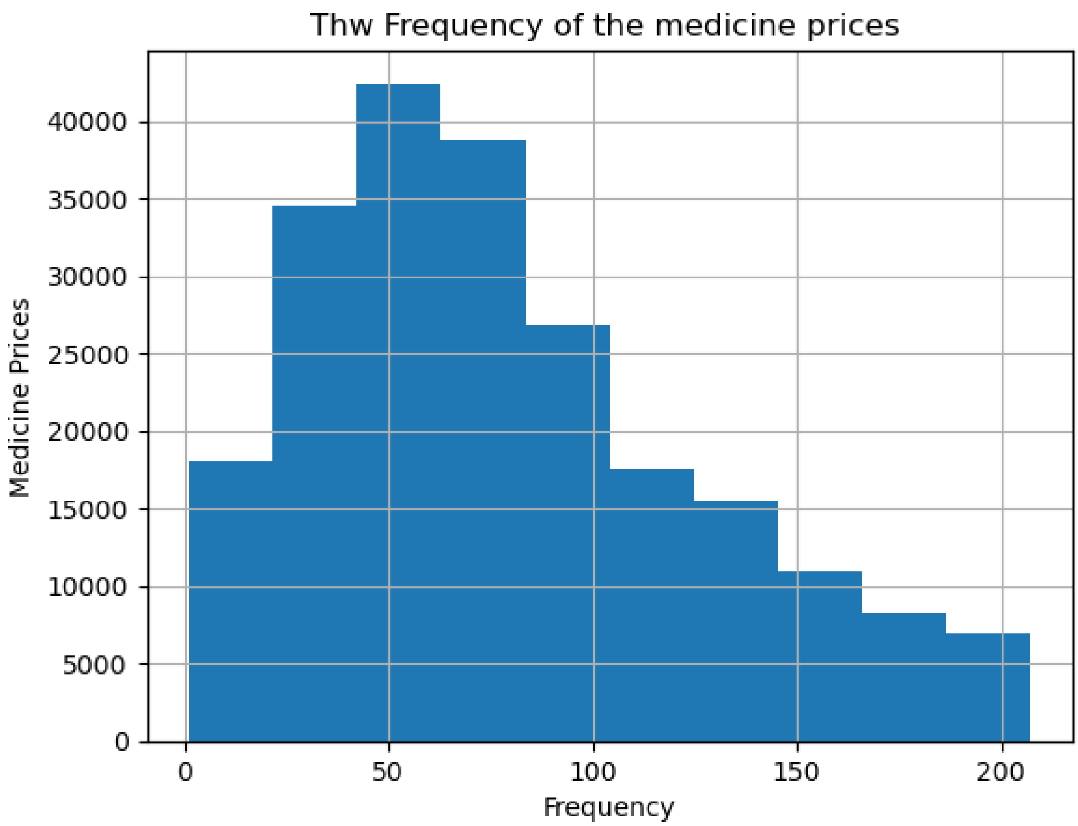
Why Histogram?

Histograms are used for numerical columns to analyze:

- Distribution
- Frequency
- Density
- Skewness

Code

```
df['price(₹)'].hist()
plt.xlabel("Frequency")
plt.ylabel("Medicine Prices")
plt.title("The Frequency of the Medicine Prices")
plt.show()
```



Observations

- Most medicines fall between ₹50–₹200.
- Distribution is highly right-skewed.
- Few medicines are extremely expensive.

Conclusion

The pharmaceutical market mainly consists of affordable medicines while specialty medicines form a small but expensive category.

8.1.2 Top 35 Most Common Values

Why Bar Plot?

Bar plots are used for categorical columns to compare category frequencies.

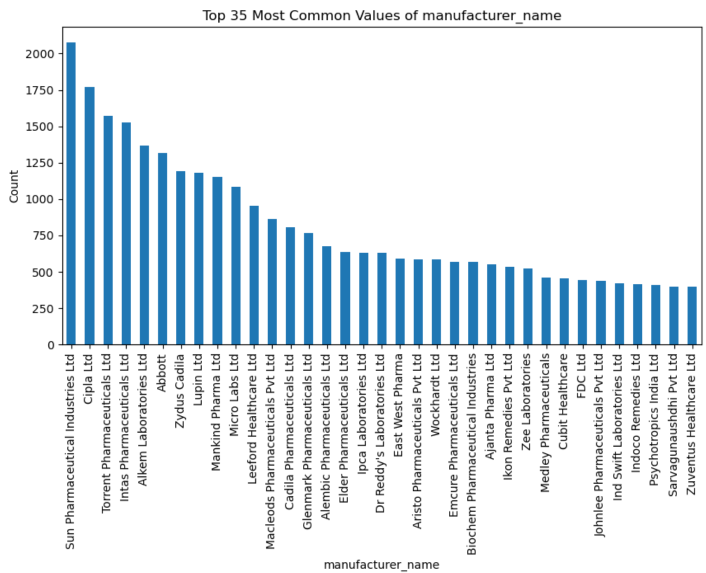
Columns Analyzed

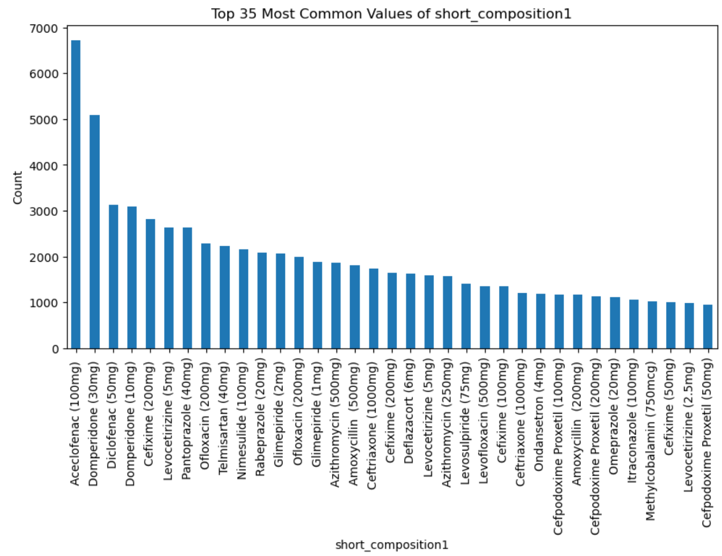
- Medicine names
- Manufacturer names
- Pack size labels

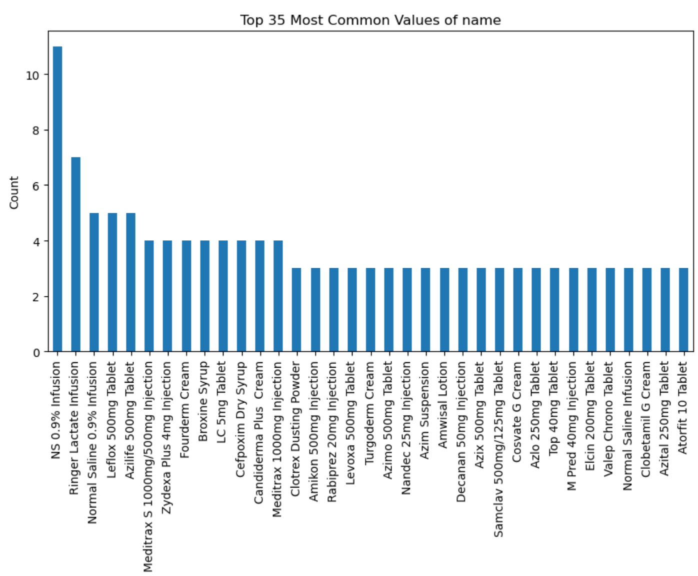
- short_composition1
- short_composition2

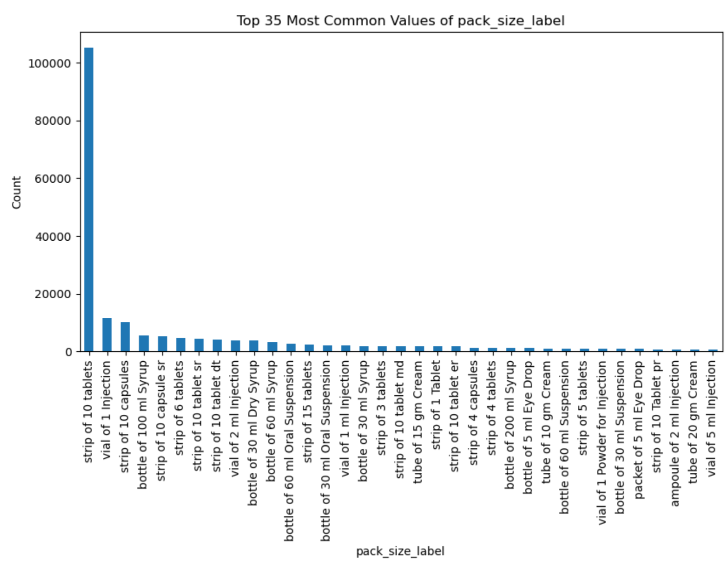
Code

```
df[col].value_counts().head(35).plot(kind='bar', figsize=(10,5))
plt.title(f"Top 35 Most Common Values of {col}")
```









Observations

- Generic medicines dominated the dataset.
- Some manufacturers appeared repeatedly.

- Tablet-related pack sizes were highly common.

Conclusion

The Indian pharmaceutical market is dominated by generic tablet medicines.

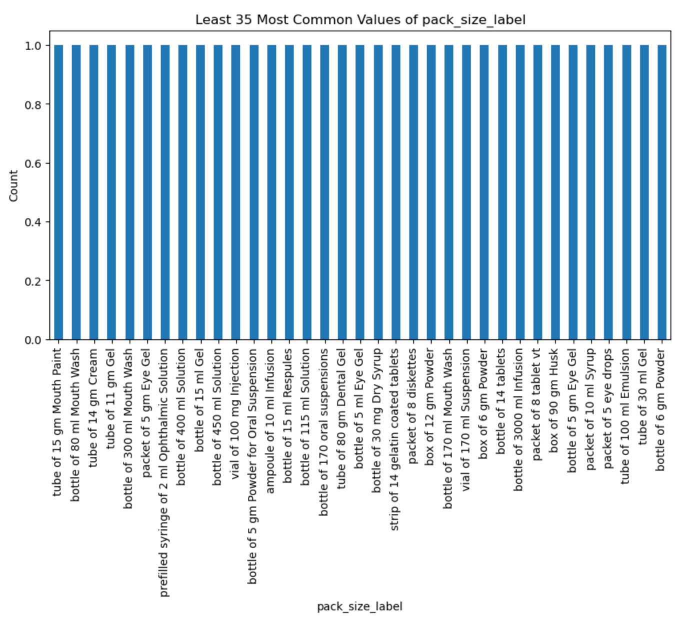
8.1.3 Least 35 Common Values

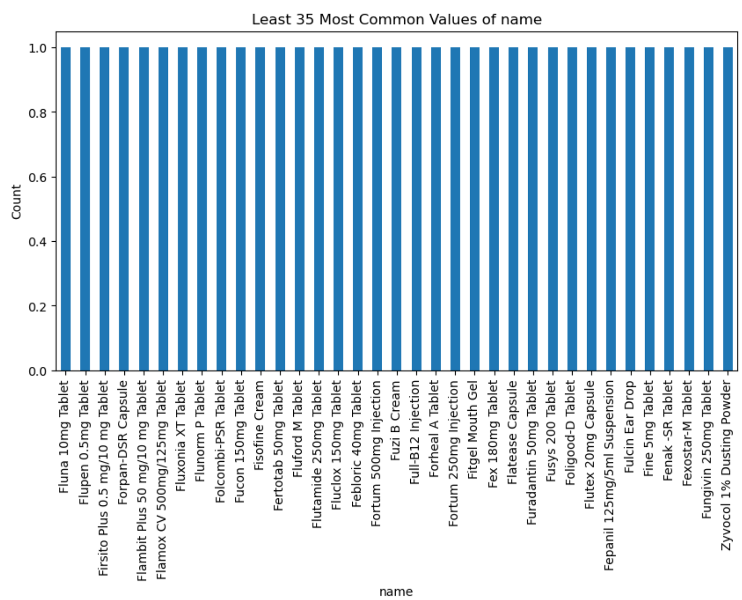
Code

```
df[col].value_counts().tail(35).plot(kind='bar', figsize=(10,5))
plt.title(f"Least 35 Most Common Values of {col}")
```

Observations

- Rare medicines appeared in smaller quantities.
- Specialty products had lower frequency.





Conclusion

Low-frequency medicines belong mainly to specialized therapeutic categories.

8.1.4 Pie Chart Analysis

Why Pie Chart?

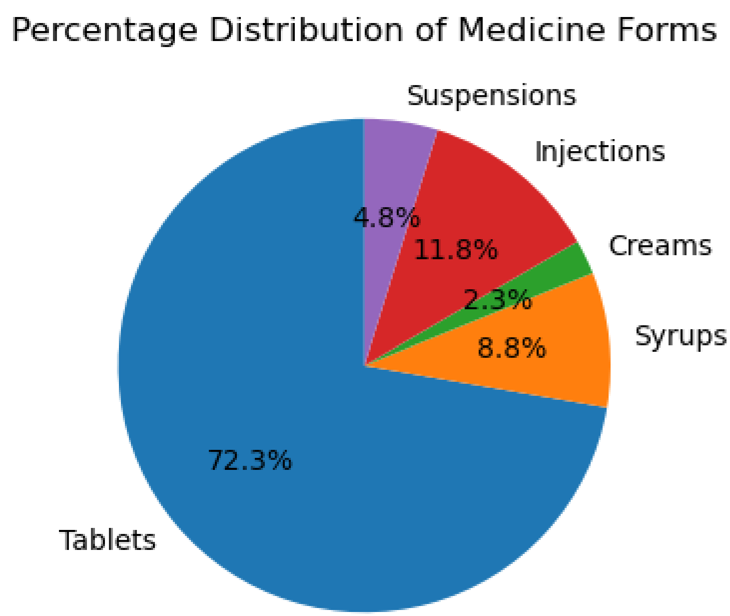
Pie charts show percentage contribution of categories.

Code

```
plt.pie(counts, labels=names, autopct='%1.1f%%', startangle=90)
```

Observations

- Tablets occupied the largest percentage.
- Syrups and suspensions occupied moderate shares.
- Injections formed a smaller portion.



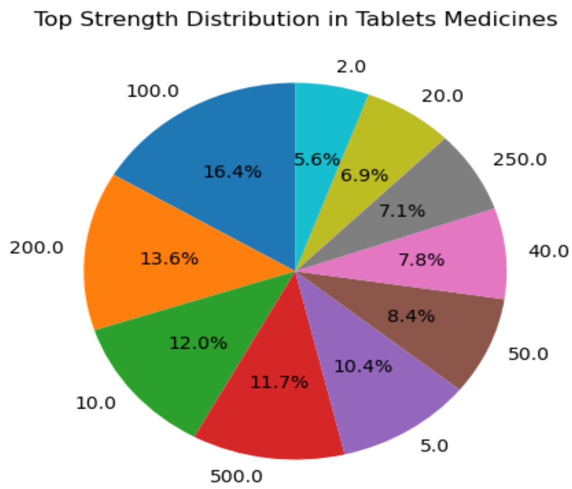
Conclusion

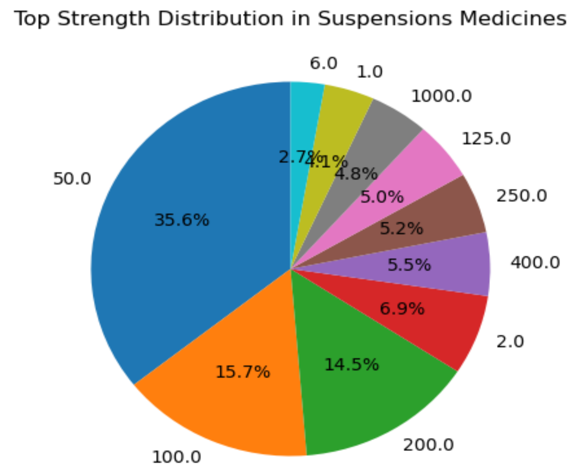
Tablet manufacturing dominates the pharmaceutical market.

8.1.5 Count of Medicine Forms

Code

```
counts = [len(data) for data in des_df]
plt.figure(figsize=(10,6))
plt.bar(names, counts)
```





Observations

- Tablets had the highest count.
- Injections had lower count but higher prices.

Conclusion

Quantity distribution and pricing distribution differ significantly.

8.2 Bivariate Analysis

Bivariate analysis studies relationships between two variables.

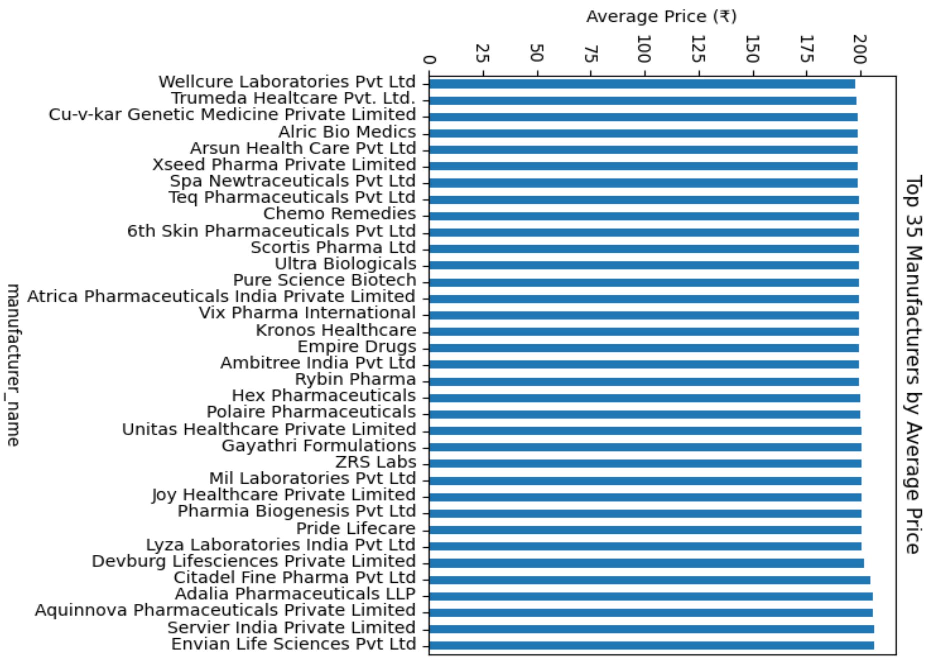
8.2.1 Average Price by Manufacturer

Code

```
df.groupby('manufacturer_name')['price(₹)'].mean().sort_values().tail(35).plot(kind='bar')
```

Observations

- Envian Life Science Pvt Ltd
- Servier India Private Ltd
- Aquinnove Pharmaceutical
- Adalia Pharmaceuticals LLP
- Citade Fine Pharma
- Devburb Lifescienc



Conclusion

Premium manufacturers focus more on specialty medicines.

8.2.2 Lowest Average Price Manufacturers

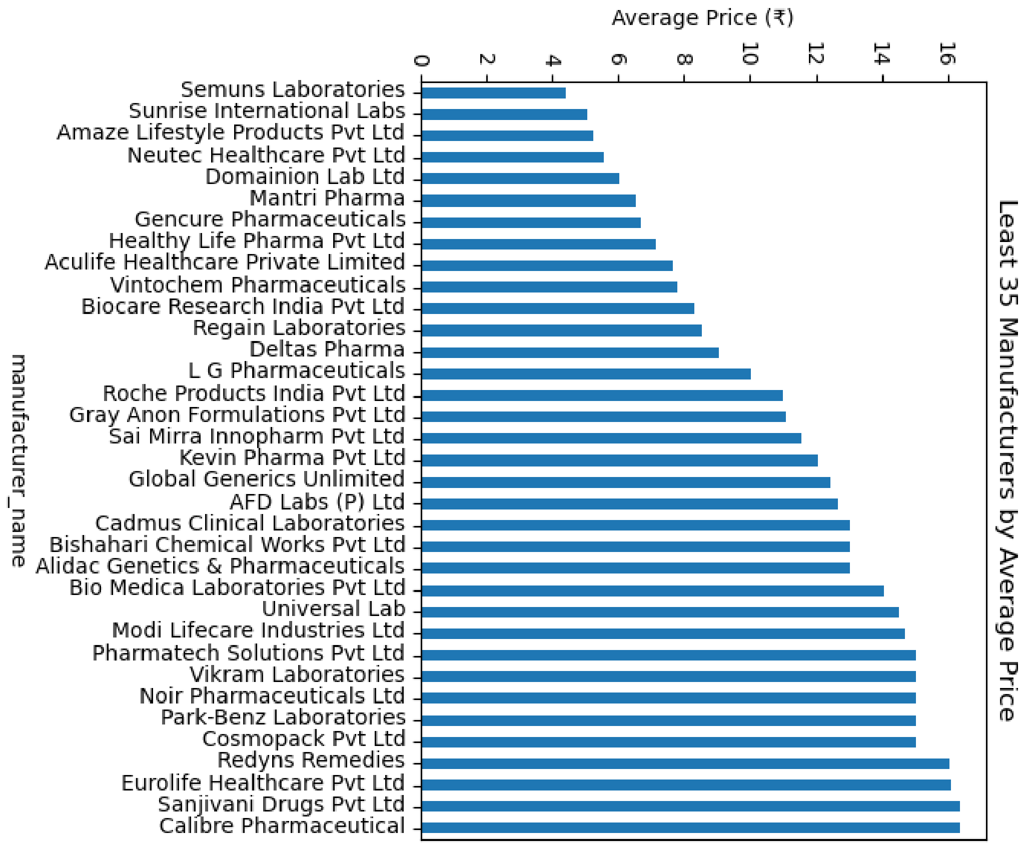
Code

df.groupby('manufacturer_name')['price(₹)'].mean().sort_values().head(35).plot(kind='bar')

Observations

- Semuns Laboratories
- Sunrise International labs
- Amaze Lifestyle Products
- Neutec Healthcare
- Domination Lab Ltd

showed lower average prices.



Conclusion

Generic manufacturers dominate affordable medicine production.

8.2.3 Statistical Description by Medicine Form

Code

```
names = ['Tablets', 'Syrups', 'Creams', 'Injections', 'Suspensions']


for i in range(len(des_df)):
    print(f"--- Description for {names[i]} ---")
    print(des_df[i]['price(₹)'].describe())
```

```
            --- Description for Tablets ---
            count    135788.000000
            mean         80.313466
            std          48.640499
```

```
min          1.050000
25%         44.000000
50%         70.000000
75%        110.000000
max        207.120000
Name: price(₹), dtype: float64

--- Description for Syrups ---
count    16545.000000
mean        67.400663
std         29.714416
min          2.500000
25%         47.300000
50%         65.000000
75%         82.000000
max        206.800000
Name: price(₹), dtype: float64

--- Description for Creams ---
count    4240.000000
mean       93.833285
std        48.839893
min         2.500000
25%        55.000000
50%        85.000000
75%       130.000000
max       207.000000
```

```
Name: price(₹), dtype: float64


--- Description for Injections ---
count    22223.000000
mean        73.835536
std         49.743703
min          1.250000
25%         33.000000
50%         62.000000
75%        110.000000
max        207.000000
Name: price(₹), dtype: float64
--- Description for Suspensions ---
count    9031.000000
mean       60.521560
std        36.003839
min         3.000000
25%        36.000000
50%        50.640000
75%        74.000000
max       207.000000
Name: price(₹), dtype: float64
```

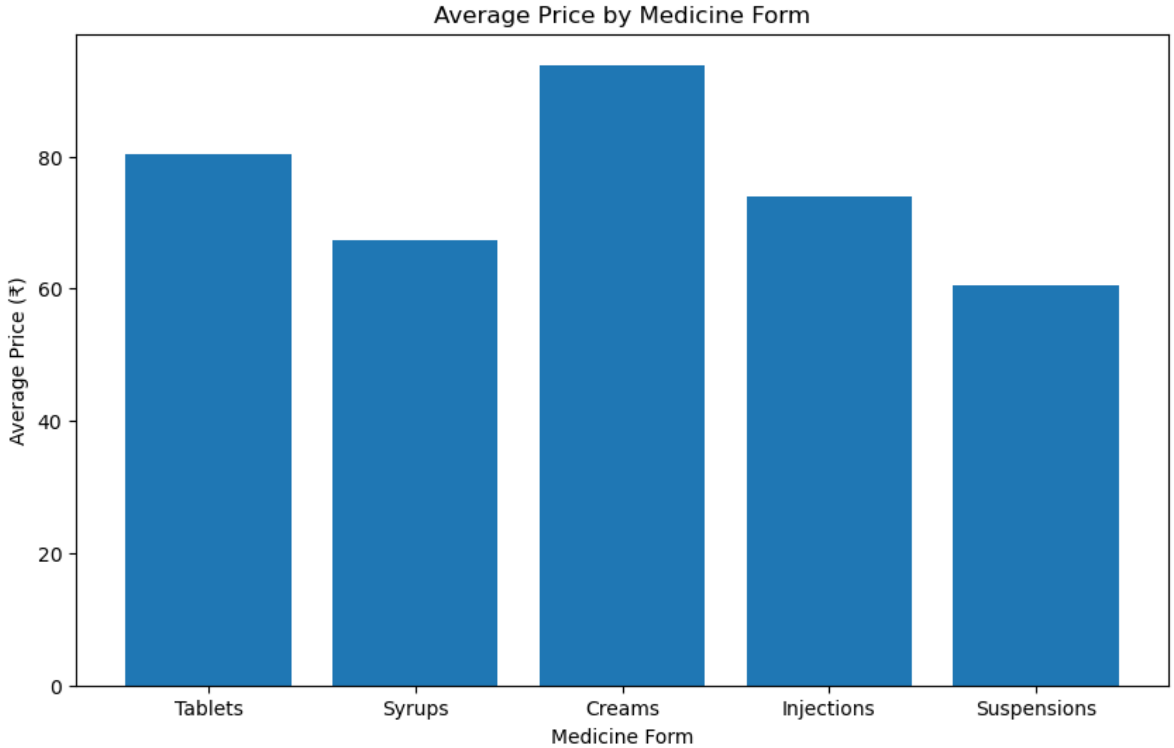
Observations

- Tablets showed lower mean prices.
- Injections showed very high maximum values.
- Creams and suspensions showed moderate variation.

Conclusion

Medicine form strongly affects pharmaceutical pricing.

8.2.4 Average Price by Medicine Form



Code

```
for data in des_df:
    avg_prices.append(data['price(₹)'].mean())


plt.figure(figsize=(10,6))
plt.bar(names, avg_prices)
```

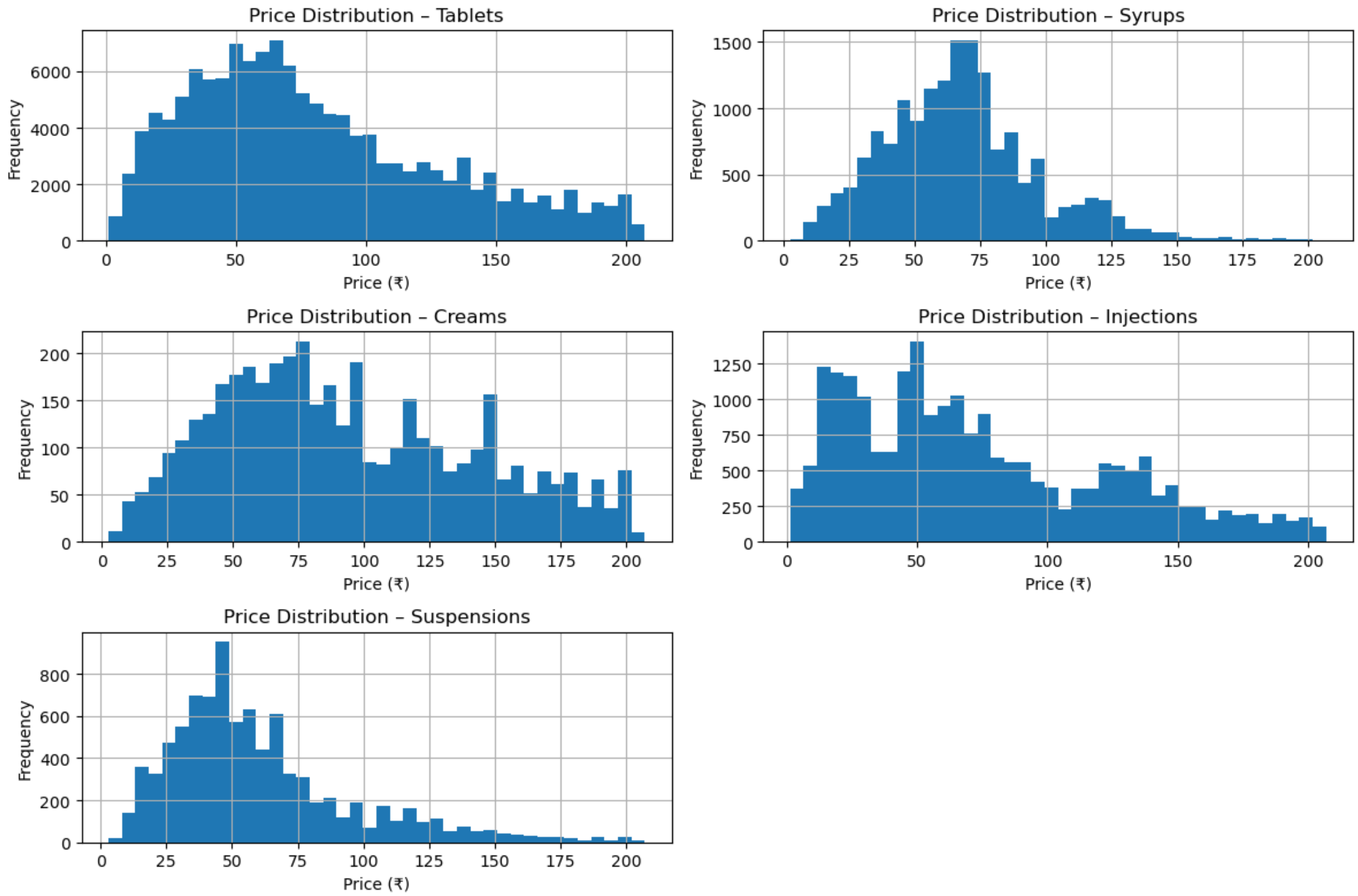
Observations

- Injections had the highest average price.
- Tablets had the lowest average price.

Conclusion

Injectables dominate premium pharmaceutical pricing.

8.2.5 Price Distribution by Medicine Form



Code

```
for i in range(len(des_df)):
    plt.subplot(3,2,i+1)
    des_df[i]['price(₹)'].hist(bins=40)
```

Observations

Tablets

- Strong cluster below ₹200.
- Stable pricing.

Syrups

- Predictable pricing range.

Creams

- Moderate price skew.

Suspensions

- More variation than syrups.

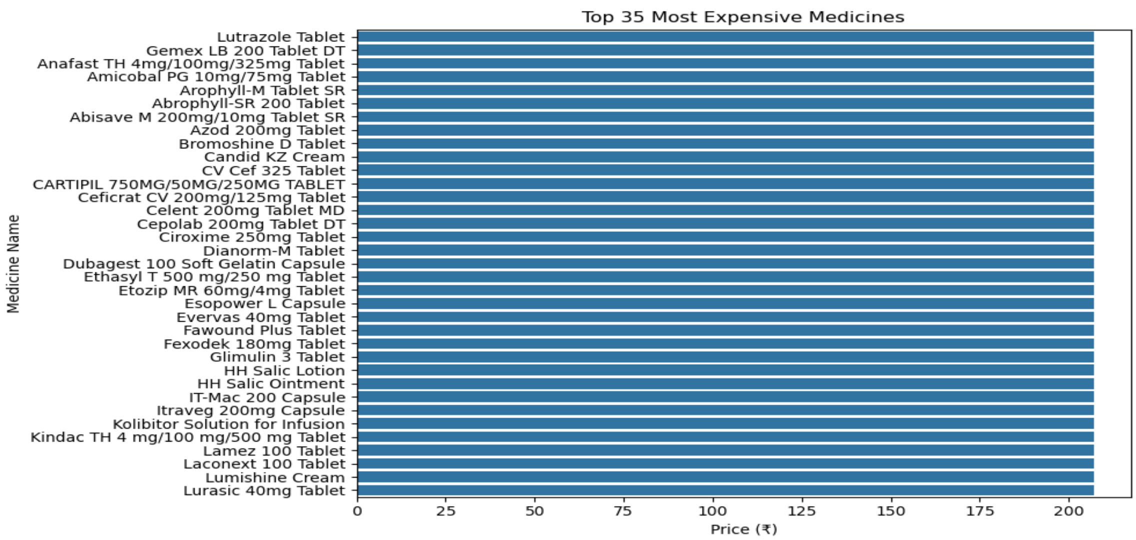
Injections

- Highest spread and strongest outliers.

Conclusion

Medicine form type is one of the strongest price determinants

8.2.6 Top 35 Most Expensive Medicines



Code

plt.figure(figsize=(10,6))

sns.barplot(x='price(₹)', y='name', data=top10_high)

Observations

- Mostly injectable medicines.
- Critical-care medicines dominated the chart.

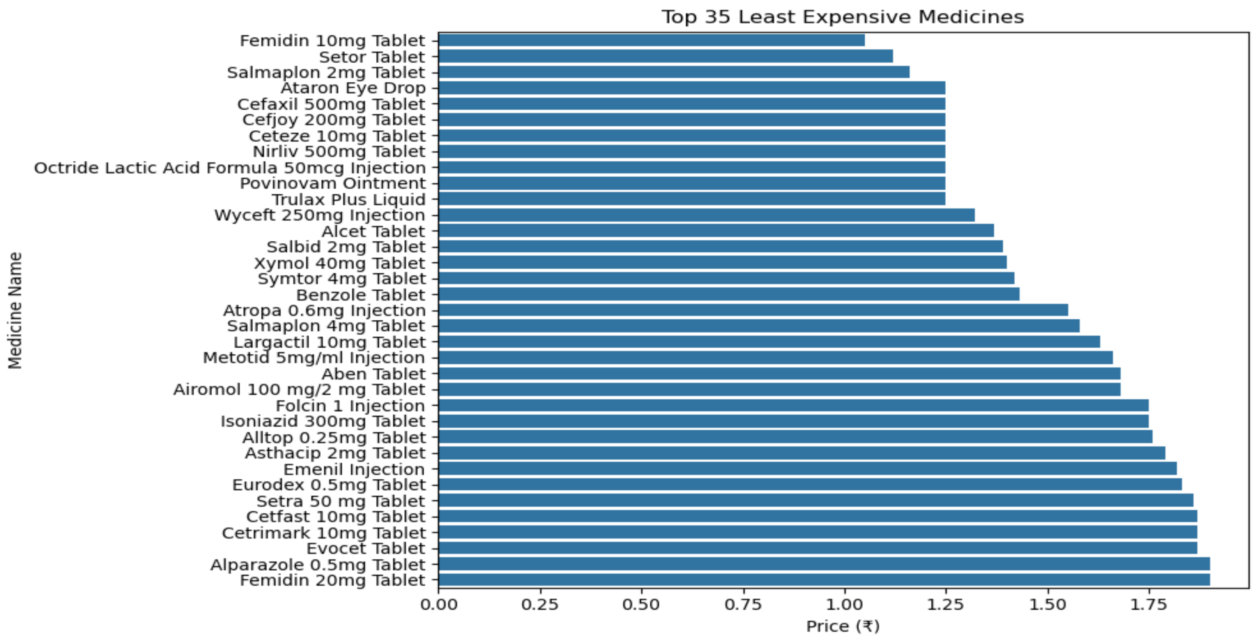
Example Medicines

- Arachitol 6L Injection
- ZI Fast Injection
- Azithral Injection

Conclusion

Specialty injections contribute heavily to pharmaceutical outliers.

8.2.7 Top 35 Least Expensive Medicines



Code

```
plt.figure(figsize=(10,6))
sns.barplot(x='price(₹)', y='name', data=top10_low)
```

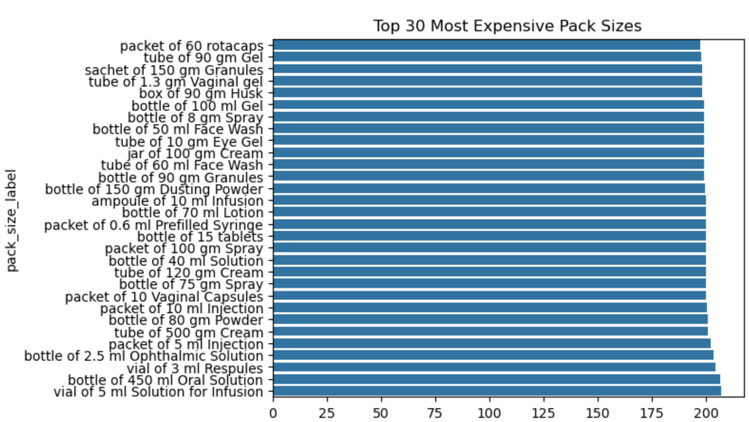
Observations

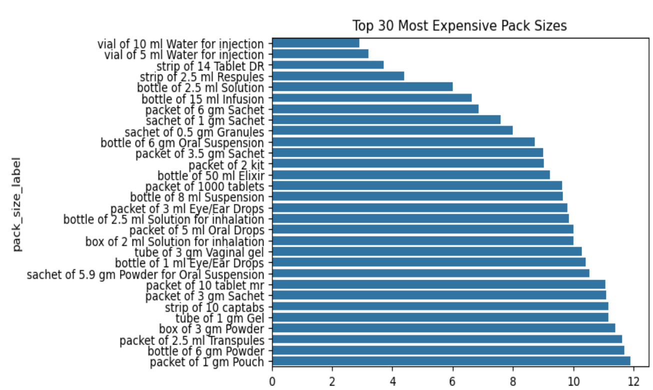
- Mostly generic tablets and syrups.
- Prices ranged between ₹1–₹15.

Conclusion

Generic medicines dominate the low-cost pharmaceutical segment.

Other bivariate analysis.





8.3 Multivariate Analysis

Multivariate analysis studies multiple variables together.

8.3.1 Violin Plot Analysis

Why Violin Plot?

Violin plots display:

- Distribution
- Density
- Spread
- Concentration

Code

```
sns.violinplot(x='form', y='price(₹)', data=combined_df)
```

Observations

Tablets

- Dense low-price cluster.

Syrups

- Stable pricing distribution.

Creams

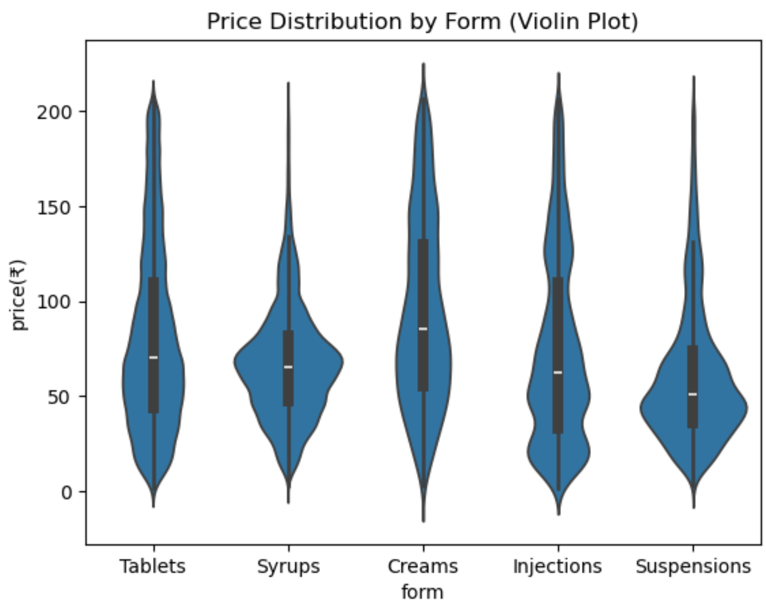
- Moderate skew.

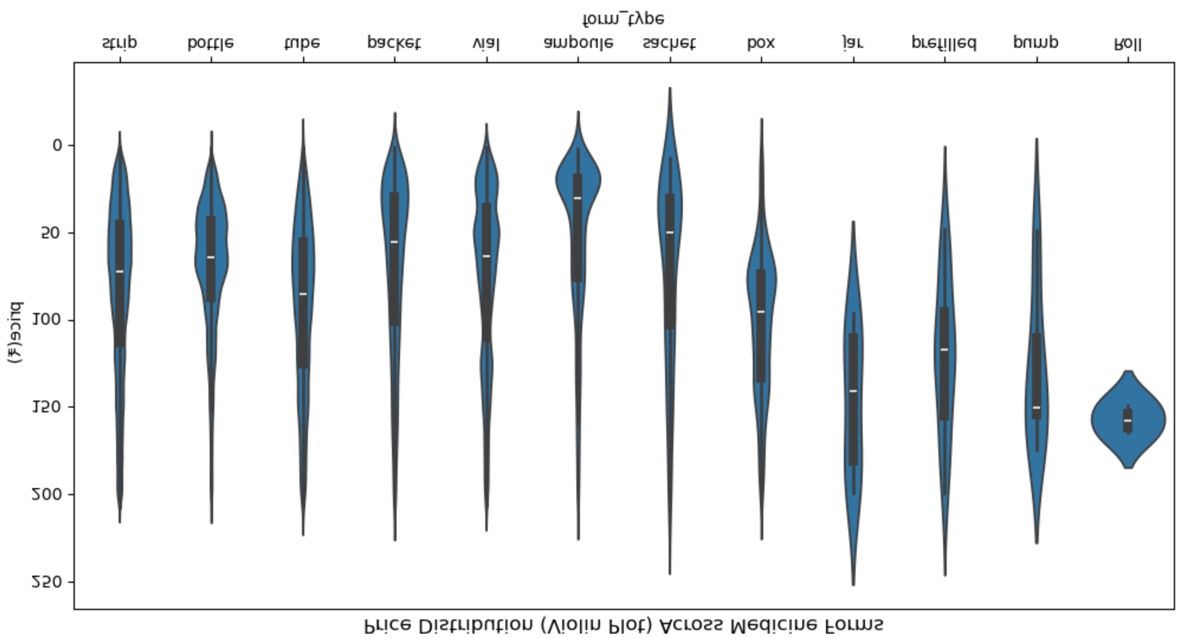
Suspensions

- Moderate spread.

Injections

- High density in upper price ranges.
- Strong outliers.





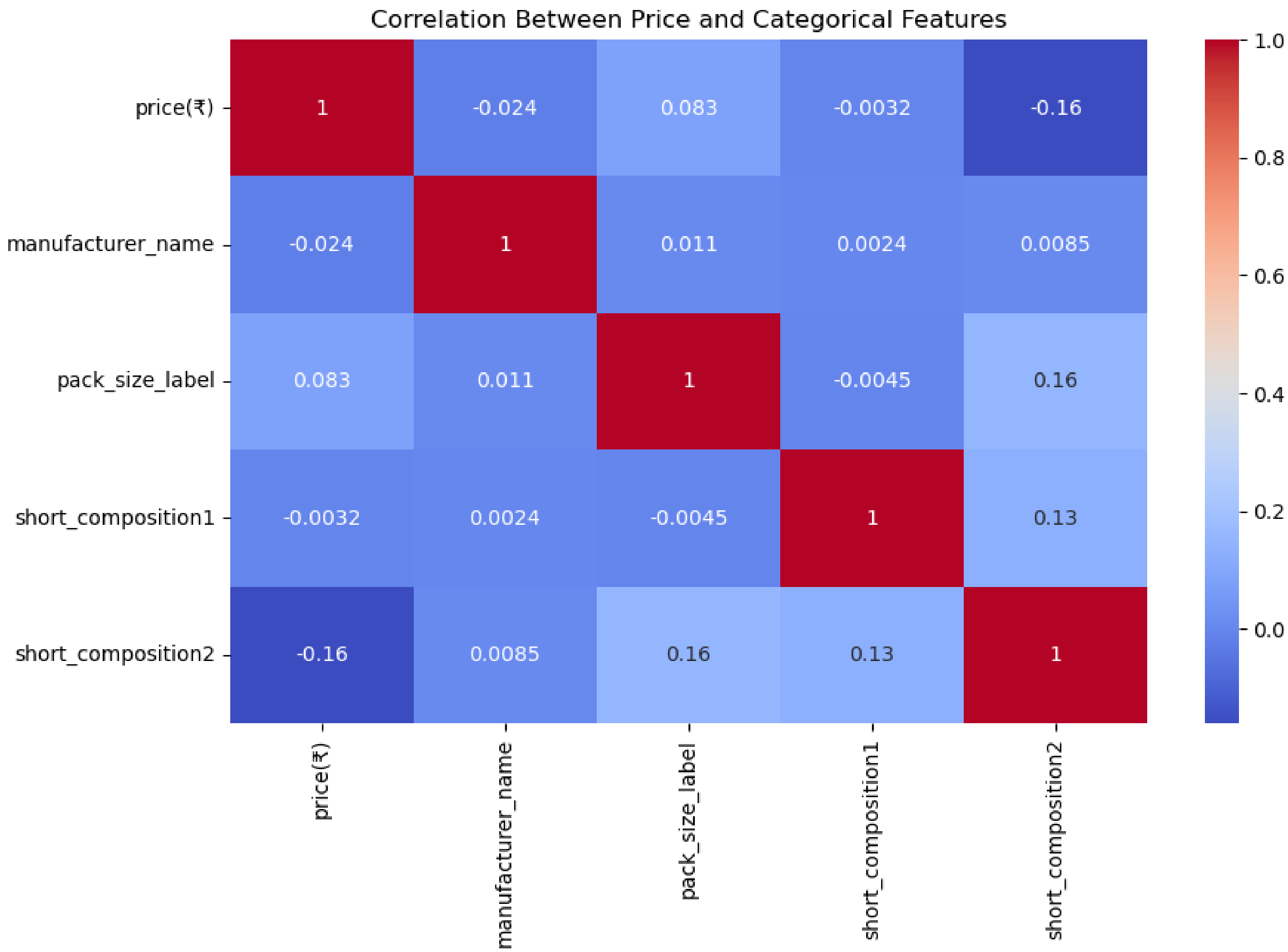
Conclusion

Violin plots clearly showed injections dominate premium pricing.

8.3.2 Heatmap and Correlation Analysis

Why Heatmap?

Heatmaps help identify correlations between features.



Observations

- Strength positively correlated with price.
- Form type and pack size affected pricing.
- Manufacturer and form combinations created price clusters.

Conclusion

Pharmaceutical pricing depends on multiple interacting variables.

Final Conclusions from Visualizations

Overall Price Distribution

- Medicine prices are highly right-skewed.
- Most medicines fall between ₹50–₹200.
- Few specialty medicines create extremely high outliers.
- The price distribution confirms that the Indian pharmaceutical market is largely dominated by affordable and mid-range medicines, while a smaller portion consists of premium therapeutic products.

Form Type Analysis

Tablets

- Most common form.
- Lowest pricing.
- Highly stable.
- Dominated the dataset in both frequency and packaging distribution.

Syrups

- Affordable and predictable pricing.
- Commonly associated with pediatric and general-use medicines.

Creams

- Moderate pricing variation.
- Slightly higher average pricing compared to syrups and suspensions after outlier removal.

Suspensions

- Wider distribution than syrups.
- Showed moderate variability due to antibiotic and specialized formulations.

Injections

- Strongest outliers.
- Highest pricing variability and widest distribution range.
- Premium therapeutic category.

- Contributed significantly to pharmaceutical price fluctuations even after outlier removal.

Manufacturer-Level Insights

Most Common Manufacturers

- Sun Pharmaceutical Industries Ltd
- Cipla Ltd
- Torrent Pharmaceuticals Ltd
- Intas Pharmaceuticals Ltd
- Alkem Laboratories Ltd
- Zydus Cadila

These companies dominated the dataset in terms of medicine count, indicating strong market presence within the Indian pharmaceutical sector.

Premium Manufacturers by Average Price

- Envian Life Sciences Pvt Ltd
- Servier India Private Limited
- Aquinnova Pharmaceuticals Private Limited
- Adalia Pharmaceuticals LLP
- Citadel Fine Pharma Pvt Ltd

Lower Average Price Manufacturers

- Semuns Laboratories
- Sunrise International Labs
- Amaze Lifestyle Products Pvt Ltd
- Neutec Healthcare Pvt Ltd
- Domination Lab Ltd

Conclusion

Manufacturer specialization and therapeutic focus strongly affect pharmaceutical pricing behavior.

Pack Size Insights

Most Common Pack Sizes

- Strip of 10 tablets
- Strip of 15 tablets
- Bottle of 100 ml syrup
- Strip of 6 tablets

These packaging formats dominated the dataset and confirmed the large-scale presence of tablet-based medicines.

Highest Price Pack Sizes

- Injection vials
- Injectable packets
- Inhalers

Lowest Price Pack Sizes

- Tablet strips
- Generic packs

Conclusion

Packaging configuration heavily influences pharmaceutical pricing and product positioning.

Strength-Based Insights

- Higher dosage strengths generally had higher prices.
- Injectable formulations showed the highest price variation.
- Combination medicines and specialized therapeutic formulations contributed to higher pricing patterns.

Final Summary

This project successfully analyzed pharmaceutical product data using data preprocessing, feature engineering, and exploratory data analysis techniques. Multiple visualizations including histograms, bar plots, pie charts, violin plots, heatmaps, and manufacturer distribution graphs were used to identify pharmaceutical pricing trends and market behavior.

The analysis revealed that medicine form, manufacturer, pack size, therapeutic specialization, and composition strength are the strongest factors influencing pharmaceutical pricing. Tablet formulations dominated the affordable medicine segment and represented the majority of products in the dataset, while injections and specialized formulations contributed heavily to premium pricing and pricing variability.

The study also highlighted the dominance of major Indian pharmaceutical manufacturers such as Sun Pharma, Cipla, Torrent Pharmaceuticals, Intas Pharmaceuticals, and Alkem Laboratories in terms of product availability and market presence.

Overall, the project provided meaningful insights into pharmaceutical pricing behavior, medicine distribution, packaging trends, and manufacturer-level market patterns within the Indian pharmaceutical industry.

The findings from this project can support:

- Healthcare analytics
- Pharmaceutical retailers
- Drug comparison systems
- Pharmaceutical market segmentation
- Medicine pricing analysis
- Pharmaceutical price optimization
- Healthcare decision-making systems

10. Challenges Faced

- Handling missing values in composition columns.
- Managing extreme outliers.
- Extracting medicine forms from text.
- Feature engineering for strength extraction.
- Visualizing large categorical datasets.

11. Future Scope

Future improvements may include:

- Machine Learning price prediction models
- Drug recommendation systems
- Brand vs generic medicine comparison
- NLP-based composition analysis
- Clustering and segmentation models
- Time-series pharmaceutical price analysis

12. References

1. OpenDataBay Pharmaceutical Dataset
2. Python Documentation
3. Pandas Documentation
4. Matplotlib Documentation
5. Seaborn Documentation
6. Scikit-learn Documentation
7. Jupyter Notebook Documentation

13. GitHub Repository

Project Notebook:
https://github.com/Tanuj-Manikanta-2007/-Pharmaceutical-Product-Data-Analysis/blob/main/G36_EDA_Pharmaceutical%20(1).ipynb

Prepared By:
Tanuj Pinninti

Project Title:
Pharmaceutical Product Data Analysis for Price Optimization & Drug Comparison